

Ontological Hierarchy of Energy Reduction The Levels of Fundamental Reality in Natural Units From Time-Mass Duality to Universal Energy Field

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Abstract

This work examines the ontological hierarchy of T0 theory under the paradigm of natural units, where through time-mass duality $T \cdot m = 1$ all physical quantities can be reduced to energy. The central insight: There exist **five ontological levels of reduction**, ranging from the most fundamental (universal energy field) to observable physics. Each level emerges from the underlying one through mathematical necessity. The analysis shows: (1) **Level 0 – Absolute Foundation**: The universal energy field $E_{\text{Field}}(x, t)$ with wave equation $\square E = 0$. (2) **Level 1 – Time-Mass Duality**: $T(x, t) \cdot m(x, t) = 1$ in natural units. (3) **Level 2 – Geometric Parameters**: $\xi = 4/30000$ and 4D torsion structure. (4) **Level 3 – Effective Field Theory**: Modified laws with $\sim 1\text{--}2\%$ corrections. (5) **Level 4 – SI Units Physics**: Classical observation level with $c, \hbar, G$ as separate constants. Narrative integration occurs through upward propagation: From the fundamental energy field emerges duality, from that geometry, from that effective laws, from that classical physics.

.1 Introduction: The Reduction Program

The Central Question

Fundamental Question

If in natural units ($\hbar = c = 1$) through time-mass duality everything can be reduced to energy, which ontological levels exist, and how do they organize themselves hierarchically?
Put differently: What are the **depths of reality** when we systematically descend from human conventions (SI units) to fundamental structures (energy field)?

The Dimensional Reduction

In natural units:

$$\hbar = c = 1 \quad \Rightarrow \quad [L] = [T] = [E^{-1}], \quad [M] = [E] \quad (1)$$

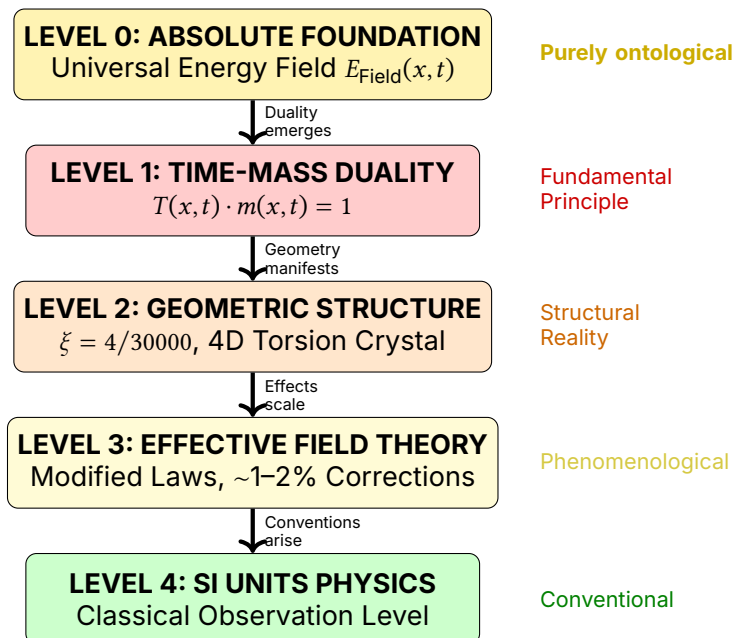
Consequence: All physical quantities are reduced to **one dimension** – energy!

Quantity	SI Units	Natural Units
Length	m	E^{-1}
Time	s	E^{-1}
Mass	kg	E
Temperature	K	E
Charge	C	dimensionless
Energy	J	E

Table 1: Dimensional reduction in natural units

.2 The Five Ontological Levels

Hierarchy Overview



.3 Level 0: The Absolute Foundation

Ontological Description

The Most Fundamental Reality

At the deepest level exists:

A Universal Energy Field $E_{\text{Field}}(x, t)$

This field is:

- **Non-dual:** No separation into space/time/mass
- **Self-evident:** Requires no further concepts
- **Dynamic:** Obeys the wave equation
- **Universal:** Permeates the entire universe

The Fundamental Equation

$$\square E_{\text{Field}}(x, t) = 0 \quad (2)$$

where $\square = \frac{\partial^2}{\partial t^2} - \nabla^2$ is the d'Alembert operator.

Physical meaning:

- Energy propagates as wave
- No sources or sinks at fundamental level
- Completely deterministic
- Local in space and time

Why is this fundamental?

Justification of Fundamentality

The energy field is fundamental because:

1. Minimal assumptions:

- Only one field
- Only one equation
- No free parameters (in natural units)

2. Maximal explanatory power:

- All other concepts emerge from it
- Space = configuration space of the field
- Time = evolution of the field
- Mass = field excitation

3. Mathematical elegance:

- Linear (superposition valid)
- Lorentz invariant
- Energy conserving

Ontological Status

What exists:

- The energy field $E_{\text{Field}}(x, t)$
- Its configuration at each time
- Its evolution dynamics
- **What doesn't exist** (at this level):
- Separate time as independent entity
- Separate mass as substance
- Particles as fundamental objects
- Space as empty container

.4 Level 1: Time-Mass Duality

Emergence of Duality

From the fundamental energy field emerges the first structuring:

Time-Mass Duality

In natural units holds the fundamental relationship:

$$T(x, t) \cdot m(x, t) = 1 \quad (3)$$

This is equivalent to:

$$T(x, t) = \frac{1}{m(x, t)} = \frac{1}{E(x, t)} \quad (4)$$

Relation to the Uncertainty Principle

The Heisenberg uncertainty principle reads:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \quad (5)$$

In natural units ($\hbar = 1$):

$$\Delta E \cdot \Delta t \geq \frac{1}{2} \quad (6)$$

This inequality does not yield an equation with the bound 1, and a limit $\Delta \rightarrow 0$ would contradict the uncertainty relation itself. Time-mass duality is therefore not derived from it but is a basic postulate:

$$E \cdot T = 1 \quad \Leftrightarrow \quad m \cdot T = 1 \quad (7)$$

The Intrinsic Time Field

The duality manifests as a field:

$$T(x, t) = \frac{1}{\max(m(x, t), \omega)} \quad (8)$$

Dimensional verification:

$$[T(x, t)] = [E^{-1}] \quad (9)$$

$$[m(x, t)] = [E] \quad (10)$$

$$[T \cdot m] = [E^{-1}] \cdot [E] = [1] \quad \checkmark \quad (11)$$

Ontological Status

At this level exist:

- Time as **field quantity** $T(x, t)$ (not as parameter)
- Mass as **field quantity** $m(x, t)$ (not as substance)
- Their reciprocal relationship as **fundamental law**

Physical meaning:

- Time varies with energy: $T \propto 1/E$
- Mass varies with energy: $m \propto E$
- Both are **aspects of the energy field**

Reduction to Energy

In natural units:

$$E = m \quad (\text{Energy} = \text{Mass}) \quad (12)$$

$$E = \omega \quad (\text{Energy} = \text{Frequency}) \quad (13)$$

$$E = 1/T \quad (\text{Energy} = \text{inverse time}) \quad (14)$$

$$E = 1/L \quad (\text{Energy} = \text{inverse length}) \quad (15)$$

Everything is energy in various manifestations!

.5 Level 2: Geometric Structure

Emergence of Geometry

From time-mass duality emerges geometric structure:

Geometric Manifestation

The duality manifests geometrically as:

- **Parameter:** $\xi = \frac{4}{30000} = 1.333 \times 10^{-4}$
- **Structure:** 4D torsion crystal
- **Scale:** Sub-Planck granulation $\Lambda_0 = \xi \cdot \ell_P$
- **Symmetry:** Pentagonal breaking via golden ratio φ

The Field Equation

The time-mass field obeys:

$$\boxed{\nabla^2 m(x, t) = 4\pi G \rho(x, t) \cdot m(x, t)} \quad (16)$$

Dimensional verification (natural units):

$$[\nabla^2 m] = [E^2] \cdot [E] = [E^3] \quad (17)$$

$$[4\pi G \rho m] = [1] \cdot [E^{-2}] \cdot [E^4] \cdot [E] = [E^3] \quad \checkmark \quad (18)$$

Geometric Parameters

From the field equation follow:

$$\beta = \frac{2Gm}{r} = \frac{2m}{r} \quad (\text{in nat. units with } G = 1) \quad (19)$$

$$\xi_{\text{geom}} = 2\sqrt{G} \cdot m = 2m \quad (\text{geometric parameter}) \quad (20)$$

The 4D Torsion Structure

Topology:

$$\mathcal{M}_{\text{fund}} = \mathbb{R}^3 \times S^1_{\text{comp}} \quad (21)$$

where:

- $\mathbb{R}^3$ = observable 3D space
- S^1_{comp} = compactified 4th dimension with radius $r_4 = \xi \cdot \ell_P$

Ontological Status

At this level exist:

- Geometric structure as **emergent property** of duality
- Parameter ξ as **manifestation** of 4D structure
- Torsion as **twisting** of compact dimension
- **Not yet existent** (only higher levels):
- Separate constants $c, \hbar, G$
- Particles as distinct objects
- Classical trajectories

.6 Level 3: Effective Field Theory

Emergence of Phenomenological Laws

From geometric structure emerge measurable effects:

Effective Description

At measurable scales ($\ell \gg \Lambda_0$) we see:

- Modified force laws with ξ -corrections
- Fractal dimension $D_f = 3 - \xi$
- Anomalous moments: no ξ deviation from the Standard Model for the electron (ξ additions to a_e excluded, Doc. 382)
- Geometric constant predictions

Modified Laws

Coulomb's law:

$$F_{\text{Coulomb}} \propto \frac{1}{r^{2-\xi}} \approx \frac{1}{r^2} \left(1 + \xi \ln \frac{r}{\ell_P} \right) \quad (22)$$

since $r^{-(2-\xi)} = r^{-2} e^{\xi \ln r} \approx r^{-2}(1 + \xi \ln r)$ (as in Doc. 145/150).

Gravitational potential:

$$\Phi(r) = -\frac{Gm}{r}(1 + \kappa r) \quad (23)$$

Fine structure constant:

$$\alpha = \xi E_0^2, \quad E_0 = 7.398 \text{ MeV} \quad \Rightarrow \quad \alpha^{-1} = \frac{1}{\xi \cdot 7.398^2} = 137.035 \quad (24)$$

The agreement with the measured value is a property of the anchor E_0 . The geometric expression $\pi^4 \sqrt{2} = 137.757$ does not match the measured value 137.035999; it lies 0.53 % above it (Doc. 149).

Correction Factors

Over many orders of magnitude, ξ accumulates:

$$K_{\text{frak}} = 1 - 100\xi \approx 0.9867 \quad (25)$$

This leads to ~1.33% corrections in observables.

Ontological Status

At this level exist:

- Effective laws as **approximations** of geometry
- Measurable deviations from Standard Model
- Phenomenological parameters (not yet $c, \hbar, G$ separate)

Characteristics:

- **Not fundamental**, but practically relevant
- **Emergent** from deeper levels
- **Approximative** with defined accuracy

.7 Level 4: SI Units Physics

Emergence of Conventions

From effective theory emerge human conventions:

Conventional Physics

For practical purposes we introduce:

- Separate constants: $c = 299\,792\,458 \text{ m/s}$, $\hbar = 1.055 \times 10^{-34} \text{ Js}$
- Separate units: Meter, kilogram, second
- Separate quantities: Energy $\neq$ mass $\neq$ time

This is the level of human measurements!

Back Translation

From natural to SI units:

$$E \text{ (nat.)} \rightarrow E \text{ (SI)} = E \cdot (\hbar c) \quad (26)$$

$$m \text{ (nat.)} \rightarrow m \text{ (SI)} = m \cdot \frac{\hbar}{c} \quad (27)$$

$$T \text{ (nat.)} \rightarrow T \text{ (SI)} = \frac{T}{c} \quad (28)$$

Ontological Status

At this level exist:

- Human conventions as **measurement tools**
- Separate concepts for practical applications
- Classical approximations for everyday physics

Characteristics:

- **Not fundamental**, but conventional
- **Useful** for technology and experiments
- **Obscures** the deeper unity of physics

.8 Hierarchy Summary

The Complete Chain

Level	Description	What exists	Status
0	Energy Field	$E_{\text{Field}}(x, t)$	Absolutely fundamental
1	Time-Mass Duality	$T \cdot m = 1$	First emergence
2	Geometry	ξ , 4D Torsion	Structural reality
3	Effective Theory	Modified Laws	Phenomenological
4	SI Physics	$c, \hbar, G$ separate	Conventional

Table 2: The five ontological levels

Causal Relationships

$$\text{Level 0} \Rightarrow \text{Level 1} \Rightarrow \text{Level 2} \Rightarrow \text{Level 3} \Rightarrow \text{Level 4} \quad (29)$$

where $\Rightarrow$ means: "determines" or "allows to emerge"

Reduction to Energy

At all levels holds in natural units:

$$[X] = [E]^n$$

for some $n \in \mathbb{Z}$

Everything is energy!

.9 Narrative Integration

Bottom-Up: The Emergence Narrative

The Story of Reality

LEVEL 0 – In the beginning was the field:

There exists a universal energy field $E_{\text{Field}}(x, t)$ that obeys the wave equation $\square E = 0$. Nothing else exists – only this one field.

⇓

LEVEL 1 – Duality emerges:

As a basic postulate, time-mass duality structures the field: $T \cdot m = 1$; it does not follow from the uncertainty principle. Time is no longer parameter, but field!

⇓

LEVEL 2 – Geometry manifests:

The duality manifests geometrically: 4D torsion crystal with parameter $\xi = 4/30000$, compact 4th dimension at sub-Planck scale.

⇓

LEVEL 3 – Effects scale:

At measurable scales we see modified laws: Coulomb $\propto 1/r^{2-\xi}$, geometric constants.

⇓

LEVEL 4 – Conventions arise:

Humans introduce SI units: meter, kilogram, second. They artificially separate $c, \hbar, G$. The deeper unity is obscured.

Top-Down: The Reduction Narrative

The Path to Fundamentality

START: SI Physics (Level 4)

We begin with separate concepts: energy, mass, time, length. We have many constants: $c, \hbar, G, k_B, \dots$

⇓ *Simplification*

Natural Units (Level 3)

We set $c = \hbar = 1$. Suddenly: energy = mass, time = inverse energy. Everything becomes simpler!

⇓ *Deeper analysis*

Geometric Structure (Level 2)

We recognize: The simplicity comes from 4D geometry. Parameter ξ encodes everything. Torsion explains mass!

⇓ *Complete reduction*

Time-Mass Duality (Level 1)

We understand: Time and mass are dual, $T \cdot m = 1$. Both are aspects of energy!

↓ *Fundamental truth*

Universal Energy Field (Level 0)

At the foundation: One field, one equation. Everything else emerges.

.10 Comparison of Both Descriptions

4D Torsion Crystal vs. Energy Reduction

4D Torsion Crystal (Level 2)	Energy Reduction (Level 0–1)
Geometric perspective Intuitive: Twisting 4 dimensions topological	Field-theoretic perspective Abstract: Duality 1 dimension (energy) reductive
Torsion as cause Sub-Planck structure primary	Field excitation as cause Wave equation primary
BOTH describe the same reality!	
Level 2 in hierarchy Emerges from Level 1 Geometrically manifest	Level 0–1 in hierarchy Fundamental for Level 2 Energetically fundamental

Table 3: Complementary descriptions

Ontological Classification

How do both fit in?

Energy Reduction (Level 0–1):

- **More fundamental** – goes deeper
- **More abstract** – less intuitive
- **More universal** – holds without restrictions

4D Torsion Crystal (Level 2):

- **Emergent** – follows from Level 1
- **More intuitive** – geometrically visualizable
- **Structural** – manifests duality

Relationship:

Energy Field (Level 0) $\xrightarrow{\text{creates}}$ Duality (Level 1) $\xrightarrow{\text{manifests}}$ Geometry (Level 2)

Why Both Descriptions Coexist

Complementarity

Analogous to wave-particle duality in quantum mechanics:

Energy Reduction:

- Like wave description
- Fundamental, but abstract
- Mathematically elegant
- Hard to visualize

4D Geometry:

- Like particle description
- Emergent, but intuitive
- Geometrically intuitive
- Practically useful

Both are valid, describing different aspects of the same reality!

.11 Practical Consequences

For Calculations

Which level to choose?

Level 0–1 (Energy Reduction):

- Theoretical derivations
- Fundamental principles
- Symmetry arguments
- Conceptual clarity

Level 2 (Geometry):

- Visual explanations
- Particle masses
- Structural predictions
- Narrative presentations

Level 3 (Effective):

- Experimental predictions
- Comparison with data
- Phenomenology

Level 4 (SI):

- Practical measurements
- Technology
- Everyday applications

For Communication

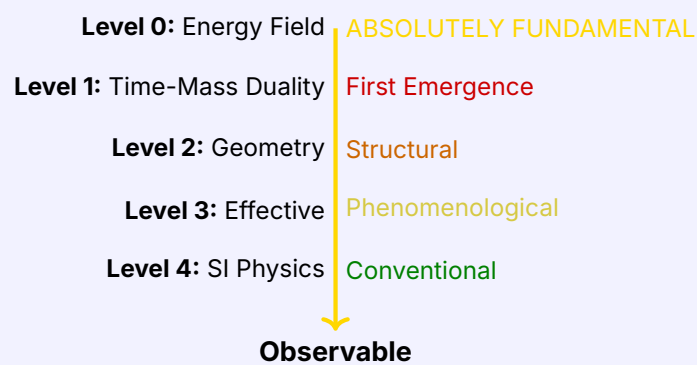
Target Audience	Preferred Level	Reason
Laypeople	Level 4 (SI)	Familiar
Students	Level 3 (Effective)	Learnable
Physicists	Level 2 (Geometry)	Intuitive
Theorists	Level 1 (Duality)	Fundamental
Philosophers	Level 0 (Field)	Ontological

Table 4: Level choice by target audience

.12 Conclusion

Main Result

T0 theory possesses a clear ****five-level ontological hierarchy****:



Through natural units, everything is reduced to energy.
 The 4D geometry is Level 2 – emergent from duality (Level 1).
 The universal energy field (Level 0) is the absolute foundation.

The Complete Reduction

The Truth of Physics

Everything is Energy

Space, time, mass, charge, forces, particles – all these are only different ****manifestations of a single universal energy field****.

In natural units this becomes mathematically explicit:

$$[X] = [E]^n \quad \text{for every physical quantity } X \quad (30)$$

The time-mass duality $T \cdot m = 1$ is the key to this insight.

The 4D torsion crystal is the geometric manifestation of this fundamental truth.